

Roman Jahandideh

Human-Computer Interaction · Bioadaptive XR · Embodied & Collaborative Systems

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portfolio

RESEARCH PROFILE

Interdisciplinary HCI researcher working across architecture, landscape architecture, and interactive arts. I design and evaluate real-time bioadaptive environments, wearable-sensing interfaces, and collaborative game systems, pairing Unity prototyping and physiological sensing with controlled user studies and research-through-design. Applying for doctoral study in Canada to begin Fall 2027.

RESEARCH INTERESTS

Generative & AI-assisted design for the built environment · Physiologically adaptive & biofeedback-driven systems · Embodied interaction in virtual & game environments · Human-centered empirical user studies

EDUCATION

Master's in Interactive Arts and Technology — Simon Fraser University, Surrey, BC 2025 – present

- Second master's degree. Research: bioadaptive game worlds, wearable physiological sensing, embodied self-regulation. Expected: _____

Master's in Landscape Architecture Design — Istanbul Technical University, Istanbul 2020 – 2024

- Virtual Landscape / Interface X Lab: virtual landscapes and depth perception in head-mounted displays.

Bachelor's in Architecture Design — Tabriz Islamic Art University, Tabriz, Iran 2011 – 2016

Relevant graduate coursework: Human-Computer Interfaces · Serious Games · Research Design · Scientific Research, Ethics & Seminar · Contemporary Art

PUBLICATIONS & MANUSCRIPTS

Jahandideh, R., et al. (2023). *Creating Immersive Virtual Landscapes: A User-Centred Approach to Enhance Depth Perception in Head-Mounted Displays*. eCAADe 2023: Digital Design Reconsidered, Vol. 1, p. 149.
doi:10.52842/conf.ecaade.2023.1.149

Jahandideh, R., et al. (2024). *Optimizing Eye-Tracking Locomotion in Virtual Environments: The Effects of Colour, Luminance, Light Intensity and Movement Speed*. Full paper accepted, CAADRIA 2024.

Jahandideh, R. (2026). *Reading a Remote Teammate's Body: A Body-Worn Sensing Interface for Group Coordination in a Distributed Interactive Space*. Under review, ACM. Icon-legibility study (n = 36) and live three-player cooperative play (12 teams).

Jahandideh, R. (2026). *Embedding Physiological Responsiveness into Game World Dynamics: Designing Bio-adaptive Game Worlds for Embodied Self-Regulation*. Submitted to ACM SIGGRAPH 2026. Within-subjects study, n = 24.

In preparation: *The Illusion of Distance (egocentric distance and object transparency in VR); Lights Out, Way On (eye-tracking locomotion in virtual landscapes); The Human Touch in AI Architecture.*

RESEARCH EXPERIENCE

Graduate Researcher — iViz Lab, School of Interactive Arts + Technology, SFU 2025 – present

- BioTerrarium: built a bio-adaptive environment in Unity 6 where live heart rate and HRV from an Apple Watch drive fog, vegetation and light, so stress clouds the world and calm restores clarity. Live prototype: biofeedbackexperiment.netlify.app
- Implemented the sensing pipeline (HealthKit and Swift bridge, Bluetooth Low Energy, WebSocket into Unity) with end-to-end latency under 200 ms and RMSSD-based zone mapping.
- Ran a counterbalanced within-subjects study (n = 24) with HR/HRV, PANAS, FSS and MAAS; paired t-tests, correlation, and reflexive thematic analysis of interviews.
- Designed an ambient heads-up cue encoding a remote teammate's heart rate and variability without numbers or emotion labels; validated online (n = 36) and in live cooperative play (12 teams).

Researcher — Interface X Lab / Virtual Landscape Lab, Istanbul Technical University 2020 – 2024

- Designed and ran VR experiments in Unreal Engine with head-mounted displays on depth and egocentric distance perception, object transparency, colour and luminance, and eye-tracking locomotion.
- Creative direction and interface graphics for seven VR research projects; taught paper structure and research writing to undergraduate design students.

VR / UX Developer — Computer and Technology Lab, Tabriz Islamic Art University

2014 – 2016

- Interactive simulations and VR used to teach spatial cognition, with 3D modelling, animation, texturing and lighting for educational content.

TEACHING

- Teaching Assistant, IAT 102 Graphic Design, SFU (Sept 2026 – present): studio and lab sessions, critique, grading.
- Teaching Assistant, IAT 343 3D Animation, SFU (May 2025 – present): Maya and Cinema 4D instruction across three semesters and 40+ students; mentored project teams through full animation pipelines.
- Teaching Assistant, IAT 313 Narrative & New Media (W-designated), SFU (May 2025 – Jan 2026): two semesters of weekly discussions, feedback on interactive work in Unity and Unreal, writing and revision support.
- Earlier instruction: Adobe Photoshop instructor (2015 – 2019) and 3ds Max / architectural visualization instructor (2016 – 2020), Urmia, Iran; workshops at Istanbul Technical University; English tutor, Cambly.

UNIVERSITY SERVICE

- Senator, SFU Senate, representative for the Faculty of Communication, Art and Technology (Mar – Jun 2025): input on teaching and research policy, course development and curriculum review.
- Member, Senate Graduate Studies Committee, SFU (Mar 2025 – present): review of new graduate programs, course and program changes, and thesis and supervisory guidelines.

AWARDS & CREATIVE PRACTICE

- Best Dissertation Award (2016) and Second Prize, Landscape Design Competition (2013), Tabriz Art University; Sixth Prize, Sustainable Cabin for Natural Disasters, Memar News (2018); Best Abstract Photography, Tabriz Art Festival (2018); NODET, Iran (2004 – 2011).
- Independent digital artist: two digital art collections released on OpenSea and Foundation (2020 – 2022); animation and visualization work in the portfolio above.

TECHNICAL SKILLS

Programming: Python (University of Michigan certificates); C# for Unity; Swift for a HealthKit / Bluetooth biosignal bridge; JavaScript with D3.js, Three.js and WebGL for interactive data visualization and web prototypes; HTML and CSS; visual scripting in Unreal Blueprints and Grasshopper.

Sensing & data: Apple Watch and HealthKit, Bluetooth Low Energy, WebSocket streaming, networked multiplayer prototypes, HR and HRV (RMSSD) processing, filtering, normalization and zone mapping, sub-200 ms feedback loops.

Real-time 3D & tools: Unity 6, Unreal Engine, Maya, Blender, Cinema 4D, 3ds Max, Rhino + Grasshopper, AutoCAD; Adobe Creative Suite, Figma, DaVinci Resolve; laser cutting, CNC milling, 3D printing.

Research methods: Within-subjects and counterbalanced design, research-through-design, PANAS / FSS / MAAS, t-tests and correlation, semi-structured interviews, reflexive thematic analysis, human-subjects ethics protocols.

Languages: Persian and Azerbaijani (native), Turkish (advanced), English (fluent; research and university teaching).

REFERENCES

Dr. Steve DiPaola — Professor, School of Interactive Arts + Technology, Simon Fraser University
sdipaola@sfu.ca · Course supervisor, IAT 343 Animation (Teaching Assistant, 2025 – present)

Susan Clements-Vivian — Associate Director and University Lecturer, School of Interactive Arts + Technology, Simon Fraser University
sclement@sfu.ca · Course supervisor, IAT 313 Narrative & New Media (Teaching Assistant, 2025 – 2026)

Dr. Ikhwan Kim — Assistant Professor, Department of Landscape Architecture, Istanbul Technical University
kimi@itu.edu.tr · iikim3@gmail.com · Research supervisor, virtual landscape and VR lab, ITU (2020 – 2024)